

Variables

Labeled dabbe jinki value badalti rehti hai

01 Opening Hook

STORY TIME

Socho tumhare paas ek labeled dabba (box) hai jispar likha hai 'x'. Aaj tumne usmein 5 seb rakhe. Kal tumne wahi dabba khali karke usmein 10 chocolates rakh diye.

Dabba wahi hai, naam wahi hai ('x'), lekin andar ki cheez badal sakti hai. Isli 'labeled dabbe' ka naam hai Variable.

02 Why Does This Concept Exist?

THE PROBLEM

Agar humein kisi unknown ya changing quantity ke saath kaam karna ho - jaise 'ghar ka size', 'model ka weight', 'temperature' - toh humein ek naam chahiye jisse hum us quantity ko refer kar sakein bina uski exact value pehle se jaane. Variable invent hua taaki hum general rules likh sakein jo kisi bhi value ke liye kaam karein, na ki sirf ek fixed number ke liye.

03 Explain Like I'm 10

SIMPLE WORDS

School mein blackboard par teacher likhti hai: 'Agar tumhare paas x chocolates hain aur main tumhe 2 aur doon, toh total kitni?' Yahan 'x' ek placeholder hai - kisi bhi number ke liye kaam karega. Agar $x=3$, answer 5. Agar $x=10$, answer 12.

AI mein bhi bilkul waise - model ke andar hazaron 'dabbe' (variables) hote hain jinke naam hote hain jaise w_1 , w_2 , b - inki values training ke dauraan badalti rehti hain jab tak model sahi answer nahi deta.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Do tarah ke variables samajhna zaroori hai: (1) Input variables – jo data se aate hain (jaise photo ke pixels), aur (2) Parameters/Weights – jo model khud seekhta hai training ke dauraan. Jab hum kehte hain 'model train ho raha hai', hum literally keh rahe hain ki 'model ke andar ke variables (weights) apni values badal rahe hain taaki output better ho'.

05 Ab Thoda Maths

FORMULA TIME

Variable Notation:

x , y , w , b jaise letters variables ko represent karte hain. Jaise: $y = wx + b$, yahan w aur b variables hain jo model seekhta hai, x input variable hai, y output hai.

Independent vs Dependent Variable:

Independent variable (x) – jo hum control/input karte hain. Dependent variable (y) – jo x par depend karta hai, uska result hai.

Constants vs Variables:

Constant kabhi nahi badalta (jaise $\pi = 3.14159$). Variable badal sakta hai – AI mein weights training ke dauraan badalte rehte hain, isliye woh variables hain, constants nahi.

06 Visualize Everything

PICTURE IT

VARIABLE = LABELED BOX JISKI VALUE BADAL SAKTI HAI

Before Training	After Training
+-----+	+-----+
w = 0.2	w = 0.91
+-----+	+-----+
(random start)	(learned value)

$$y = w * x + b$$

			+--- bias (variable, learned)
		+-----	input (variable, from data)
	+-----	weight (variable, learned)	
+-----	output (depends on above)		

07 AI Connection

WHERE AI USES THIS

Neural Networks: Har connection (edge) ke beech ek weight variable hota hai. GPT-jaise models mein billions of weight variables hote hain jo training se seekhe jaate hain.

Gradient Descent: Training loop mein har step par weight variables ki value thodi thodi update hoti hai taaki loss kam ho.

Hyperparameters vs Parameters: Parameters (weights) model khud seekhta hai; Hyperparameters (learning rate, batch size) hum manually set karte hain - dono variables hain lekin control karne wala alag hai.

08 Real Industry Example

REAL COMPANIES

Meta (Llama models): Llama models mein weight variables ki sankhya (parameter count) hi model ka size batati hai, jaise 'Llama 70B' matlab 70 billion weight variables.

NVIDIA: GPU training ke dauraan har second millions of weight variables ko parallelly update karte hain - isliye specialized hardware chahiye hota hai.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein jab tum `nn.Linear(10, 5)` banate ho, PyTorch internally weight aur bias variables (tensors) create kar deta hai jinki value random se start hoti hai. `requires_grad=True` set hone par PyTorch track karta hai ki training ke dauraan yeh variables kaise change honi chahiye. `optimizer.step()` call karne par yeh variables actually update hote hain.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Variable sirf ek single number hota hai.

Sahi baat: AI mein variable ek poora tensor (multi-dimensional array of numbers) ho sakta hai, na ki sirf ek number.

Galat soch: Training ke baad variables fix ho jaate hain forever.

Sahi baat: Fine-tuning ya continued training mein wahi weight variables phir se update ho sakte hain - woh permanently frozen nahi hote jab tak explicitly freeze na kiya jaaye.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Variable aur constant mein kya farak hai?

A: Variable ki value badal sakti hai, constant ki value fix rehti hai.

[Intermediate] Q: Model 'parameters' aur 'hyperparameters' mein kya farak hai?

A: Parameters (weights, bias) model training ke dauraan khud seekhta hai; hyperparameters (learning rate, epochs) developer manually pehle se set karta hai.

[Advanced] Q: Kyun weight variables ko random values se initialize kiya jaata hai, zero se nahi?

A: Agar saare weights zero se start honge, toh saare neurons same gradient seekhenge (symmetry problem) aur network diverse features nahi seekh payega - isliye random initialization zaroori hai.

12 Exercises

PRACTICE TIME

[Easy] $y = 3x + 5$ mein agar $x = 2$ hai toh y kya hoga?

Solution: $y = 3(2) + 5 = 11$.

[Medium] Ek model mein $w=0.5$, $x=4$, $b=1$. $y = wx+b$ nikalo.

Solution: $y = 0.5*4 + 1 = 3$.

[Hard] Agar training ke pehle sabhi weights same value (0.5) se initialize kiye jaayein, toh kya problem ho sakti hai?

Solution: Symmetry problem - sabhi neurons same gradient paayenge aur same tarah update honge, isliye network diverse patterns seekh nahi payega.

13 Golden Summary

Concept	Easy Meaning
Variable	Ek labeled quantity jiski value badal sakti hai
Parameter/Weight	Model ke andar ka variable jo training se seekha jaata hai
Hyperparameter	Developer dwara manually set kiya gaya variable (jaise learning rate)
Independent Variable	Input jo hum control karte hain
Dependent Variable	Output jo input par depend karta hai

14 One-Line Memory Trick

“Variable ek labeled dabba hai – naam wahi rehta hai, andar ki value badalti rehti hai.”